The training advisor: a specification of `advisor.py`

Describes `src/training/advisor.py` against exercise dictionary version 10: 31 movements, 16 cells. October 2026.

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1 Scope

The advisor is a ranking function, not a program generator. Asked on a given day, it answers *what is most needed*: which movements, in which style, have been neglected relative to a target rate. How many pieces to do, and what the session looks like, stay with the human.

It has two queries. `advise()` ranks candidate movements for one part of a session (Section 7); `recommend()` says which style is most needed overall (Section 8).

What the scorer reads. From each logged set, exactly three fields: the date, the style, and the movement. Reps, load, distance, duration, workout structure, free-text description and result are stored for the human's own record and never reach the model.

Time. Time is measured in whole calendar days. All rates below are per day unless stated as per week.

2 Axes

2.1 Patterns

The set of **patterns** is

$$\mathcal{P} = \{\text{squat}, \text{hinge}, \text{push}, \text{pull}, \text{flexion}, \text{brace}, \text{engine}\}.$$

Each is defined by an archetype, which is what the movement weights in Section 2.3 are judged against.

pattern	archetype
squat	back squat
hinge	deadlift
push	strict press
pull	strict pull-up
flexion	toes-to-bar, V-up
brace	resisting movement under load: carries, planks
engine	sustained monostructural output: running, rowing, biking

2.2 Styles

The set of **styles** is $\mathcal{S} = \{\text{light}, \text{heavy}, \text{hhr}\}$. A style is how a movement was trained, chosen per movement when logging.

style	meaning
light	inside a workout, unloaded or light
heavy	classic strength work: fresh, long rests
hhr	heavy at high heart rate: roughly 60–70% of 1RM, for reps, while breathing hard

2.3 Movements and the matrix E

The dictionary (`exercises.json`) lists the set of **movements** $\mathcal{X}$, currently 31 plus the placeholder `rest`. Each movement x carries two authored attributes:

- a row of **pattern weights** $E_{x,p} \geq 0$ with $\sum_p E_{x,p} = 1$;
- a set of **admissible styles** $A(x) \subseteq \mathcal{S}$.

The weights *split* a movement between patterns and never duplicate it: a thruster is $E = 0.5$ squat $+0.5$ push, not 1 of each. So one logged movement always contributes total weight 1, however many patterns it touches. `rest` has an empty row ($E_{\text{rest},p} = 0$ for all p); it lets interval work be written as movement rows and contributes nothing.

E and A are authored judgment, not fitted from data.

2.4 Cells

Definition 1. Let $D \subseteq \mathcal{S}$ be the styles the user has switched off (Section 10; normally empty). The set of **cells** is

$$\mathcal{C} = \big\{(p, s) \in \mathcal{P} \times \mathcal{S} \ : \ \exists x \in \mathcal{X} \text{ with } E_{x,p} > 0,\ s \in A(x),\ s \notin D\big\}.$$

A cell is a pattern–style combination that some movement can actually train. The cells are *derived* from the dictionary, not declared: adding a movement can create a cell as a side effect, and the model will then expect that cell to be trained. Of the $7 \times 3 = 21$ combinations, 16 are cells. The table gives, for each cell, the number of movements that can serve it ($E_{x,p} > 0$ and $s \in A(x)$); a dash means the combination is not a cell.

pattern	light	heavy	hhr
squat	1	6	7
hinge	2	3	5
push	2	7	9
pull	2	2	3
flexion	1	–	–
brace	–	3	4
engine	7	–	–

> **Open.** Two cells, squat/light (air squat) and flexion/light (toes-to-bar), are served by a single movement each. Whenever such a cell is suggested, it is always the same movement. A test guards against cells with *no* movement; nothing guards against cells with one.

3 From the log to increments

3.1 Events

The log is a list of sets. The scorer reduces it to a set of **events**

$$V = \big\{(t, s, x) : \text{some set of movement } x \text{ was logged at style } s \text{ on day } t\big\}.$$

Because V is a set, repeats collapse: five sets of back squats on one day are one event, and a unilateral movement logged left and right is one event.

3.2 Increments

Each event feeds weight into cells. Let $\eta = \texttt{HHR_CREDIT} = 0.5$. An event (t, s, x) produces, for every pattern p with $E_{x,p} > 0$:

1. an increment of weight $E_{x,p}$ to cell (p, s) on day t;
2. if $s = \text{hhr}$, a further increment of weight $\eta\, E_{x,p}$ to cell (p, heavy) on day t.

Write I_c for the list of increments (t_i, w_i) received by cell c. Increments addressed to a combination that is not a cell are dropped. Two different movements trained on the same day both contribute to a cell they share.

The second rule is the **hhr credit**: loaded conditioning partly, but never fully, substitutes for fresh heavy work. It counts toward the heavy cell's rate only. It does not count as having trained heavy for the "days since this cell was last trained" figure the advisor reports alongside each suggestion.

4 The exponentially weighted rate

4.1 Target rates and memory

Each style has a target rate in exposures per week,

$$r_{\text{light}} = 1, \qquad r_{\text{heavy}} = 0.75, \qquad r_{\text{hhr}} = 0.25,$$

and each pattern a weight ω_p, currently 1 for every pattern. The target rate of cell (p, s) in exposures per day is

$$\rho_{p,s} = \frac{r_s\, \omega_p}{7}.$$

A single dimensionless constant $k = \texttt{MEMORY_K} = 2$ sets each cell's memory, its **time constant**

$$\tau_c = \frac{k}{\rho_c}.$$

Memory is therefore measured in units of the cell's own target interval: a cell meant to be trained rarely remembers for longer. With all $\omega_p = 1$:

style	r_s (per week)	target interval $1/\rho$ (days)	τ (days)
light	1.00	7.0	14.0
heavy	0.75	9.3	18.7
hhr	0.25	28.0	56.0

4.2 The EWMA

Definition 2. The **exponentially weighted moving average** of cell c on day t is

$$m_c(t) = \sum_{(t_i, w_i) \in I_c,\ t_i \le t} w_i\, e^{-(t - t_i)/\tau_c}.$$

The code computes the same quantity by the recursion $m \leftarrow m\, e^{-\Delta/\tau} + w$ over increments in date order, then decays from the last increment to t. The whole history is replayed; there is no window. Old training matters less only because it has decayed.

Proposition 1. *$\hat{\rho}_c(t) = m_c(t)/\tau_c$ is an estimate of the cell's recent exposure rate: under a constant flow of ρ weight per day, $m_c \to \rho\,\tau_c$, so $\hat{\rho}_c \to \rho$.*

Proof. In continuous time, $m(t) = \int_0^t \rho\, e^{-(t-u)/\tau}\, du = \rho\tau(1 - e^{-t/\tau}) \to \rho\tau$. □

In particular a cell trained exactly at its target rate settles at $m_c = \rho_c\tau_c = k$, whatever its target. This is what puts every cell on one scale.

5 The score

Definition 3. The **score** of cell c on day t is

$$\sigma_c(t) = -\log \frac{m_c(t)}{k}, \qquad \sigma_c(t) = +\infty \text{ if } m_c(t) = 0.$$

Positive means behind target, negative means ahead.

By Proposition 1, $m_c/k = \hat{\rho}_c/\rho_c$, so

$$\boxed{\sigma_c = \log \frac{\rho_c}{\hat{\rho}_c}}$$

The score is the log-ratio of target rate to recent actual rate. Everything below follows from this reading.

(i) **Steady fractions.** Training a cell steadily at a fraction f of its target gives $\sigma = -\log f$. Half the target reads $\log 2 \approx 0.69$ behind; double reads 0.69 ahead.

(ii) **Neglect grows linearly.** With no new increments, $m_c(t + \Delta) = m_c(t)\, e^{-\Delta/\tau_c}$, so

$$\sigma_c(t + \Delta) = \sigma_c(t) + \frac{\Delta}{\tau_c}.$$

An untouched cell's score rises by $1/\tau_c$ per day, without bound: 0.071 per day for light, 0.054 for heavy, 0.018 for hhr. Growing by $\log 2$ takes $\tau_c \log 2$ days: 9.7, 12.9 and 38.8 days respectively.

(iii) **Never trained.** A cell with no increments scores $+\infty$. There is no principled finite ceiling for "never done", and ranking and sorting handle $+\infty$ correctly.

5.1 Training exactly on target

Real training arrives in discrete events, not as a flow. Suppose a cell receives weight 1 every $T = 1/\rho$ days, forever. Let $q = e^{-T/\tau} = e^{-1/k}$, the decay over one interval.

Proposition 2. *In the periodic steady state:*

(a) just after an exposure, $m^+ = \dfrac{1}{1-q}$; just before the next, $m^- = \dfrac{q}{1-q}$;

(b) the time-average of m over an interval is exactly k;

(c) the score is a sawtooth rising linearly from $\sigma^+ = \log\big(k(1-q)\big)$ to $\sigma^+ + 1/k$, with time-average

$$\bar{\sigma} = \log\big(k(1 - e^{-1/k})\big) + \frac{1}{2k} = \frac{1}{24k^2} + O(k^{-3}).$$

Proof. (a) The fixed point of $m^+ = q\,m^+ + 1$. (b) $\frac{1}{T}\int_0^T m^+ e^{-u/\tau}\,du = \frac{\tau}{T}\,m^+(1-q) = \frac{\tau}{T} = k$. (c) Between exposures the score grows at rate $1/\tau$ (item (ii) above) for T days, a rise of $T/\tau = 1/k$; its average is the midpoint $\sigma^+ + 1/(2k)$. The expansion follows from $k(1-e^{-1/k}) = 1 - \frac{1}{2k} + \frac{1}{6k^2} - \dots$ □

At $k = 2$: the score swings between -0.240 just after training and $+0.260$ just before the next session is due, and averages $+0.010$. Training exactly on target reads ahead immediately after a session and behind just before the next, and is on target on average. The swing has width $1/k$; a larger k narrows it at the price of slower-reacting scores.

6 Choosing a movement for a cell

6.1 Stalest eligible movement

The movements **eligible** for cell (p, s) are

$$X_{p,s} = \{x \in \mathcal{X} : E_{x,p} > 0,\ s \in A(x)\}.$$

The **staleness** of a movement on day t is the number of days since it was last logged at *any* style, or $+\infty$ if never. The cell's suggested movement is the stalest eligible one; ties are broken by the larger $E_{x,p}$, then by the identifier (the lexicographically last wins, which is arbitrary but deterministic). Within a cell, this is what rotates movements; the score itself is a property of the cell, not of the movement.

6.2 The session rule

Let Q be the movements already picked in the current session, and

$$L(Q) = \{p : E_{x,p} > 0 \text{ for some } x \in Q\}$$

the patterns they touch. Once a pattern is in $L(Q)$, *no* cell of that pattern is offered again in the session, at any style. This rule is categorical: it excludes cells, it does not lower their scores.

7 `advise()`: ranking a session

`advise(mode, chosen)` is called with mode = heavy, which considers the heavy cells, or mode = workout, which considers the light and hhr cells together; `chosen` is Q. It returns, for every cell of those styles not excluded by the session rule, the pair (stalest eligible movement, score), sorted by score, highest first. Ties keep the order in which cells are listed: light before hhr, then pattern alphabetically.

Picking a movement does not change any score, since nothing has been logged yet; it only adds to Q and so removes that movement's patterns from every later list. The ranking is greedy: it never looks ahead to what a pick forecloses, because the number of pieces is the human's choice. The Today page shows the result as three columns, heavy, light and hhr.

8 `recommend()`: which style is most needed

For each style s that has cells, let

$$M_s = \text{median}\{\sigma_c : c = (p, s) \in \mathcal{C}\}.$$

The median rather than the mean, because a single never-trained cell scores $+\infty$ and would decide a mean on its own.

Rank the styles by M_s. Let M_1 be the largest, M_2 the second largest (0 if there is only one style). The verdict is:

1. *no strong signal* if $M_1 \le 0$: every style is on or ahead of target, and being least ahead is not being needed;
2. *no strong signal* if $M_1 = M_2 = +\infty$: nothing has been trained in either style's typical cell, which says nothing about which is more needed;
3. otherwise, with margin $\Delta = M_1 - M_2$: *clear* if $\Delta > \log 2$, *leaning* if $\Delta > \frac{1}{2}\log 2$, and *no strong signal* otherwise.

Since scores are log-ratios of rates, $\Delta > \log 2$ says the top style's typical cell is more than twice as far below target, as a ratio, as the second style's.

> **Open.** The thresholds $\log 2$ and $\frac{1}{2}\log 2$ are free choices. $\log 2$ has a meaning (half the target rate), but which multiples of it count as "clear" and "leaning" is not derived.

9 Capacity

The weekly **demand** is the total target rate over all cells, $\sum_{(p,s)\in\mathcal{C}} r_s\,\omega_p$. With 6 light cells, 5 heavy and 5 hhr:

$$\underbrace{6\times 1}_{\text{light}} + \underbrace{5\times 0.75}_{\text{heavy}} + \underbrace{5\times 0.25}_{\text{hhr}} = 6 + 3.75 + 1.25 = 11.0 \text{ per week.}$$

Every logged movement supplies total weight 1 to the cells of its style (Section 2.3), and an hhr movement supplies a further $\eta = 0.5$ to the heavy cells. Three sessions a week of one heavy piece and four workout pieces is 15 pieces: 3 to the heavy cells directly, 12 to the light and hhr cells.

> **Open.** The design notes give supply as "roughly 12" pieces a week, with no derivation. Counted as above it is 15 plus the hhr credit. The comparison that matters is per style: heavy demand is 3.75 a week against 3 heavy pieces, so the heavy cells stay on target only with help from the hhr credit, which the closed-loop simulation shows they get.

10 Per-user settings

Every quantity above is read from an `AdvisorConfig`, so each user can be scored against their own targets. The settings page offers three knobs, translated by `user_config.py`:

setting	effect	consequence
volume multiplier v	$r_s \leftarrow v\,r_s$ for every style	$\tau_c \leftarrow \tau_c / v$
conditioning focus c	$\omega_{\text{engine}} \leftarrow c$	engine cells' target and memory rescale
disabled styles D	removes those styles' cells	they are never suggested or compared

η and k are not exposed as settings.

11 A worked example

The full dictionary, and this log, evaluated on 15 January 2026:

day	movement	style	logged as
1 Jan	back squat	heavy	5 sets
5 Jan	thruster	hhr	3 sets
8 Jan	air squat	light	3 sets
12 Jan	back squat	heavy	5 sets

Events and increments. Sixteen sets collapse to four events. Back squat is $E = 1$ squat; thruster is 0.5 squat +0.5 push; air squat is 1 squat. The increments are:

cell	increments (t_i, w_i)	τ
squat/heavy	(1 Jan, 1), (5 Jan, 0.25), (12 Jan, 1)	18.67
push/heavy	(5 Jan, 0.25)	18.67
squat/hhr	(5 Jan, 0.5)	56
push/hhr	(5 Jan, 0.5)	56
squat/light	(8 Jan, 1)	14

The two 0.25 increments to the heavy cells are the hhr credit: $\eta \times 0.5$ from the thruster.

EWMAs and scores. For squat/heavy, the increments are 14, 10 and 3 days old:

$$m = 1 \cdot e^{-14/18.67} + 0.25\, e^{-10/18.67} + 1 \cdot e^{-3/18.67} = 0.4724 + 0.1463 + 0.8515 = 1.4702,$$

$$\sigma = -\log(1.4702/2) = 0.3077.$$

The others follow in the same way:

cell	m	σ
push/heavy	0.1463	2.6152
squat/hhr	0.4182	1.5649
push/hhr	0.4182	1.5649
squat/light	0.6065	1.1931
squat/heavy	1.4702	0.3077

The other 11 cells have never been trained and score $+\infty$.

Squat/heavy reads 0.31 behind even though back squats came twice in two weeks, a little above the target of 0.75 a week. The log starts on 1 January, so the EWMA started from zero two weeks ago; it has not yet accumulated the history a steady trainee would have. See Section 12.

`advise(heavy).` The three never-trained heavy cells come first, in cell order, then the two trained ones:

cell	suggested movement	score
brace/heavy	suitcase carry	$+\infty$
hinge/heavy	deadlift	$+\infty$
pull/heavy	strict pull-up	$+\infty$
push/heavy	strict press	2.6152
squat/heavy	front squat	0.3077

Front squat, not back squat, is suggested for squat/heavy: it has never been logged, so it is the stalest eligible squat movement.

The session rule. After picking deadlift, `advise(workout)` no longer offers hinge/light or hinge/hhr: hinge is in $L(Q)$. Every other workout cell is unchanged.

`recommend()`. At least half of each style's cells are untrained, so $M_{\text{light}} = M_{\text{heavy}} = M_{\text{hhr}} = +\infty$, and the verdict is *no strong signal* by the second guard in Section 8.

Every number in this section is recomputed from the code by

`tests/test_math_worked_example.py`.

12 Open questions

> **Open. No guarantee by a date.** A neglected cell's score rises by $1/\tau$ a day without bound (Section 5), so it eventually outranks everything. Nothing forces it to the top by any particular day: an hhr cell sitting at 0 takes 39 days to reach $\log 2$. This is a deliberate bet, to be checked against real logging.

> **Open. Start-up transient.** Every EWMA starts at zero when logging starts, so scores read behind until about τ days of history exist: two weeks for light cells, eight weeks for hhr cells. Section 11 shows it on squat/heavy. Nothing corrects for it.

> **Open. $k = 2$ is a free choice.** It sets both how fast scores react ($\tau = k/\rho$) and the width $1/k$ of the on-target sawtooth (Proposition 2).

> **Open. The hhr rule has exceptions.** A movement admissible at hhr is meant to be admissible at heavy too, since hhr means a heavy movement at reduced load. Three movements break this on purpose (kettlebell swing, devil's press, man maker): dumbbell and kettlebell conditioning movements have no heavy form anyone trains. It is a guideline for authoring entries, not an invariant.

The single-movement cells (Section 2.3), the `recommend()` thresholds (Section 8) and the supply estimate (Section 9) are flagged where they arise.